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Inventor(s)	Decottignies; Robert Irwin et al.

Modular multi-chamber processing tool having link chamber for ultra high vacuum processes

Abstract

Embodiments of link chamber for use in multi-chamber processing tools or systems are provided herein. In some embodiments, a link chamber for use in a multi-chamber processing tool includes: a link chamber body having a plurality of facets extending between a bottom plate and a top plate, wherein at least seven of the plurality of facets have a chamber opening to form a plurality of chamber openings, wherein the plurality of chamber openings are sized to pass a substrate therethrough, and wherein each of the plurality of chamber openings are configured to be coupled to a slit valve, a load lock chamber, a cover plate, a process chamber, or a second link chamber body.

Inventors: Decottignies; Robert Irwin (San Carlos, CA), Radko; Marek W (Santa Clara, CA)

Applicant: Applied Materials, Inc. (Santa Clara, CA)

Family ID: 1000008779421

Assignee: APPLIED MATERIALS, INC. (Santa Clara, CA)

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Primary Examiner: Pence; Jethro M.

Attorney, Agent or Firm: Moser Taboada

Background/Summary

FIELD

(1) Embodiments of the present disclosure generally relate to substrate processing equipment.

BACKGROUND

(2) Manufacturing of semiconductor devices typically involves performing a sequence of procedures with respect to a substrate or “wafer” such as a silicon substrate, a glass plate, etc. These steps may include polishing, deposition, etching, photolithography, heat treatment, and the like. Usually, a number of different processing steps may be performed in a single processing system or “tool” which includes a plurality of processing chambers. However, other processes may be performed at other processing locations within a fabrication facility, thereby necessitating that substrates be transported within the fabrication facility from one processing location to another. Depending on the type of semiconductor device to be manufactured, there may be a relatively large number of processing steps performed at many different processing locations within the fabrication facility.

(3) Substrates are conventionally transported from one processing location to another within substrate carriers such as sealed pods, cassettes, containers and so forth. Such transport of substrates typically involves exposing the substrates to room air, or at least to non-vacuum conditions. Either may expose the substrates to an undesirable environment (e.g., oxidizing species) and/or contaminants.

(4) Accordingly, the inventors have provided embodiments of improved multi chambered processing tools for minimizing exposure of substrates to undesirable environments when transferring substrates between processing systems.

SUMMARY

(5) Embodiments of link chamber for use in multi-chamber processing tools are provided herein. In some embodiments, a link chamber for use in a multi-chamber processing tool includes: a link chamber body having a plurality of facets extending between a bottom plate and a top plate, wherein at least seven of the plurality of facets have a chamber opening to form a plurality of chamber openings, wherein the plurality of chamber openings are sized to pass a substrate therethrough, and wherein each of the plurality of chamber openings are configured to be coupled to a slit valve, a load lock chamber, a cover plate, a process chamber, or a second link chamber body.

(6) In some embodiments, a multi-chamber processing tool includes: a factory interface (FI) having one or more loadports configured to receive a substrate; a link chamber coupled to the FI, wherein the link chamber includes a plurality of facets extending between a bottom plate and a top plate, wherein at least seven of the plurality of facets have a chamber opening to form a plurality of chamber openings, wherein the bottom plate includes one or more pump ports, and wherein the bottom plate includes a plurality of valve openings proximate corresponding ones of the chamber openings; a plurality of process chambers coupled to the link chamber via respective ones of the chamber openings; one or more load lock chambers coupled to the link chamber via respective one or more chamber openings; and a link robot disposed in the link chamber and configured to facilitate transferring the substrate from the FI to the plurality of process chambers via the one or more load lock chambers.

(7) In some embodiments, a multi-chamber processing tool, includes: a factory interface (FI) having one or more loadports configured to receive a substrate; a link chamber coupled to the FI, wherein the link chamber includes a plurality of facets extending between a bottom plate and a top plate, wherein at least seven of the plurality of facets have a chamber opening to form a plurality of chamber openings that are similarly sized, wherein the bottom plate includes one or more pump ports, and wherein the bottom plate includes a plurality of valve openings proximate corresponding ones of the chamber openings; a plurality of process chambers coupled to the link chamber via respective ones of the chamber openings; one or more load lock chambers coupled to the link chamber via respective one or more chamber openings; and a link robot disposed in the link chamber and configured to facilitate transferring the substrate from the FI to the plurality of process chambers via the one or more load lock chambers.

(8) Other and further embodiments of the present disclosure are described below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Embodiments of the present disclosure, briefly summarized above and discussed in greater detail below, can be understood by reference to the illustrative embodiments of the disclosure depicted in the appended drawings. However, the appended drawings illustrate only typical embodiments of the disclosure and are therefore not to be considered limiting of scope, for the disclosure may admit to other equally effective embodiments.
- (2) FIG. 1 depicts a schematic top view of a multi-chamber processing tool having a link chamber in accordance with at least some embodiments of the present disclosure.
- (3) FIG. 2 depicts a schematic top view of a multi-chamber processing tool having a link chamber in accordance with at least some embodiments of the present disclosure.
- (4) FIG. 3 depicts a schematic isometric view of a link chamber in accordance with at least some embodiments of the present disclosure.
- (5) FIG. 4 depicts an isometric view of a link chamber body in accordance with at least some embodiments of the present disclosure.
- (6) FIG. 5 depicts a schematic top view of multiple connected multi-chamber processing tools having link chambers in accordance with at least some embodiments of the present disclosure.
- (7) FIG. 6 depicts a schematic top view of a multi-chamber processing tool coupled to a second processing tool in accordance with at least some embodiments of the present disclosure.
- (8) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. The figures are not drawn to scale and may be simplified for clarity. Elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation.

DETAILED DESCRIPTION

- (9) Embodiments of link chambers for use in multi-chamber processing tools are provided herein. The link chamber includes a plurality of facets for modular and customizable coupling with a factory interface and a plurality of process chambers to form the multi-chamber processing tool. The link chambers provided herein are generally configurable to facilitate coupling to a plurality of process chambers such as deposition chambers, etch chambers, clean chamber, degas chamber, metrology chambers, or the any other suitable chamber for ultra-high vacuum processing. As such, the link chambers described herein advantageously provide a central hub for passing substrates from the factory interface to the plurality of process chambers via the link chamber. The link chambers are configured to provide a smaller footprint and lower cost than typical multi-chamber architectures.
- (10) The link chambers may be used to support systems with various degrees of complexity. The link chambers provided herein are also advantageously configured to be coupled with one or more other link chambers or one or more other multi-chamber processing tools via transfer vias disposed therebetween to support more complex substrate processing while minimizing floorspace and processing downtime. As such, the link chambers provided herein facilitate forming a multi-chamber processing tool or a link between multiple multi-chamber processing tool for any desired substrate process. In some embodiments, the desired substrate process may be an ultra-high vacuum metallization process, for example, a copper interconnect process. However, any suitable process requiring a vacuum or ultra-high vacuum environment may be performed with the tools provided herein.
- (11) FIG. 1 depicts a schematic top view of a multi-chamber processing tool having a link chamber in accordance with at least some embodiments of the present disclosure. The multi-chamber processing tool **100** includes a plurality of facets **114**. In some embodiments, the plurality of facets **114** comprises at least seven facets. Each of the facets **114** have a chamber opening (see plurality of

chamber openings **304** in FIG. 3) that is connectable or coupled to a load lock chamber **130**, one of a plurality of process chambers **110** (plurality of process chambers **110A-110F** shown in FIG. 1), or a transfer via (see transfer via **508** in FIG. 5). The link chamber **104** includes a transfer robot **106** disposed therein. In some embodiments, the transfer robot **106** is operable to transfer one or more substrates **112** between the load lock chamber **130** and the plurality of process chambers **110**, between the plurality of process chambers **110**. In some embodiments, the transfer robot **106** is configured to transfer the one or more substrates **112** between the plurality of process chambers **110** and the transfer via **508**.

(12) The multi-chamber processing tool **100** includes a factory interface (FI) **120** having one or more loadports **122** configured to receive one or more substrates **112**. The link chamber **104** is coupled to the FI **120** via the one or more load lock chambers **130**. The one or more substrates **112** may be transferred from the FI **120** to the one or more load lock chambers **130** via a FI robot **124** disposed in the FI **120**. In some embodiments, the one or more load lock chambers **130** include one load lock chamber as depicted in FIG. 1. In some embodiments, the one or more load lock chambers **130** include a substrate transfer mechanism configured to move the one or more substrates **112** from a location proximate the FI **120** to a location proximate the link chamber **104**. The one or more load lock chambers **130** are generally vacuum chambers coupled to respective vacuum pumps that facilitate transfer of the one or more substrates **112** from ambient pressure of the FI **120** to the vacuum pressure of the link chamber **104**. The one or more load lock chambers **130** may be coupled to respective vacuum pumps.

(13) In some embodiments, the substrate transfer mechanism is a substrate carousel **136** as depicted in FIG. 1. The substrate carousel **136** may generally comprise one or more substrate supports **134** configured to rotate within the load lock chamber to move the one or more substrates **112**. In some embodiments, the one or more substrate supports **134** are coupled to a base plate **132**. In some embodiments, the substrate carousel **136** comprises three substrate supports that rotate about a rotational axis **138**.

(14) In some embodiments, the plurality of process chambers **110** include six process chambers. In some embodiments, the plurality of process chambers **110** include four deposition chambers, for example process chambers **110A-110D**, and two cleaning chambers, for example process chambers **110E-110F**. In some embodiments, the four deposition chambers are physical vapor deposition (PVD) chambers. In some embodiments, the four deposition chambers are any suitable combination of PVD chambers, chemical vapor deposition (CVD) chambers, or atomic layer deposition (ALD) chambers. In some embodiments, the plurality of process chambers **110** are vacuum chambers coupled to respective vacuum pumps.

(15) FIG. 2 depicts a schematic top view of a multi-chamber processing tool **100** having a link chamber **104** in accordance with at least some embodiments of the present disclosure. In some embodiments, as shown in FIG. 2, the one or more load lock chambers **130** comprises two load lock chambers coupled to respective facets of the plurality of facets **114** at a first end and the FI **120** at a second end. In some embodiments, the plurality of process chambers **110** include four process chambers. In some embodiments, the substrate transfer mechanism comprises a load lock transfer robot **210**. In some embodiments, the substrate transfer mechanism comprises a servo-controlled shutter **220**. In some embodiments, the servo-controller shutter **220** comprises a shutter **224** coupled to a servo motor **226**. The servo motor **226** may be coupled to the shutter **224** via an arm **228** such that rotation of the servo motor **226** moves the shutter **224** towards or away from the link chamber **104**.

(16) Referring to FIGS. 1 and 2, in some embodiments, slit valves **115** are disposed at the interfaces of the plurality of process chambers **110** and the link chamber **104** and interfaces of the load lock chamber **130** and the link chamber **104** and configured to selectively seal one or more of the plurality of facets **114**. In some embodiments, one or more of the slit valves **115** include a linear motion actuator to move a slit valve door to selectively seal or open a slit opening to facilitate

transfer of the one or more substrates **112** therethrough. In some embodiments, one or more of the slit valves **115** include an “L” motion actuator to move the slit valve door in an “L” motion (vertical and horizontal movement) to selectively seal or open a slit opening to facilitate transfer of the one or more substrates **112** therethrough.

(17) With the slit valves **115**, the link chamber **104** can be separately isolated from the plurality of process chambers **110** and the one or more load lock chambers **130**. Vacuum pumps **160**, such as cryopumps, turbopumps, or the like, may be coupled to the link chamber **104** via pump ports **108** for pumping down the link chamber **104**. The vacuum pumps **160** are operable to maintain the vacuum level of the link chamber **104**. In some embodiments, the link chamber **104** includes two pump ports **108**, one pump port coupled to a cryopump, and a second pump port coupled to a turbopump.

(18) In some embodiments, the link chamber **104** includes a controller **150** coupled to the link chamber **104** and operable to receive data from the plurality of process chambers **110** and to provide instructions to the multi-chamber processing tool **100** to process commands directed to processing and transfer of the one or more substrates **112**. The controller **150** comprises a centralized server that may implement predictive modeling, such as machine learning, artificial intelligence, and/or deep learning, to increase throughput and efficiency of the multi-chamber processing tool **100** and any tools or systems linked to the multi-chamber processing tool **100**, for example a second multi-chamber processing tool having a second link chamber (e.g., second multi-chamber processing tool **502**) or a second processing tool (e.g., second processing tool **602**). In some embodiments, the controller **150** is configured to be in communication with a second controller **550** of the second multi-chamber processing tool **502** or second controller **650** of the second processing tool **602**. The controller **150** is operable to receive and/or send data from the second controllers **550**, **650**, apply predictive modeling to the data, and provide instructions to the second controllers **550**, **650** corresponding to process commands directed to processing in and transfer of one or more substrates from the process chambers of linked first multi-chamber processing tool **100** and the second processing tools **502**, **602**.

(19) The controller **150** generally includes a central processing unit (CPU) **152** comprising one or more processors, a memory **154**, and a support circuit **156**. The CPU **152** may be any form of a general-purpose computer processor that can be used in an industrial setting. The support circuit **156** is conventionally coupled to the CPU **152** and may comprise a cache, clock circuits, input/output subsystems, power supplies, and the like. Software routines, such as a method as described herein may be stored in the memory **154** and, when executed by the CPU **152**, transform the CPU **152** into a specific purpose computer (controller **150**). The software routines may also be stored and/or executed by a second controller (not shown) that is located remotely from the multi-chamber processing tool **100**.

(20) The memory **154** may be in the form of a non-transitory computer-readable storage media that contains instructions, when executed by the CPU **152**, to facilitate the operation of the multi-chamber processing tool **100**. The instructions in the memory **154** are in the form of a program product such as a program that implements methods of the present principles. The program code may conform to any one of a number of different programming languages. In one example, the disclosure may be implemented as a program product stored on a computer-readable storage media for use with a computer system. The program(s) of the program product define functions of the aspects (including the methods described herein). Illustrative computer-readable storage media include, but are not limited to: non-writable storage media (e.g., read-only memory devices within a computer such as CD-ROM disks readable by a CD-ROM drive, flash memory, ROM chips, or any type of solid-state non-volatile semiconductor memory) on which information is permanently stored; and writable storage media (e.g., floppy disks within a diskette drive or hard-disk drive or any type of solid-state random access semiconductor memory) on which alterable information is stored. Such computer-readable storage media, when carrying computer-readable instructions that

direct the functions of the methods described herein, are aspects of the present principles.

(21) FIG. 3 depicts a schematic isometric view of a link chamber body **310** in accordance with at least some embodiments of the present disclosure. The link chamber **104** generally includes a link chamber body **310** having the plurality of facets **114** extending between a bottom plate **306** and a top plate **308**. In some embodiments, the link chamber body **310** has an octagon shape. In some embodiments, the controller **150** is coupled to the link chamber body **310** on a side **312** of the link chamber body **310** that does not include one of the plurality of facets **114**. Each of the plurality of facets **114** include a chamber opening to form a plurality of chamber openings **304**. The plurality of chamber openings **304** are sized to pass a substrate therethrough. In some embodiments, the plurality of chamber openings **304** are similarly sized.

(22) Each of the plurality of chamber openings **304** are configured to be coupled to a slit valve (e.g., slit valve **115**), a load lock chamber (e.g., load lock chamber **130**), a process chamber (e.g., process chamber **110**), a cover plate (e.g., cover plate **410**), or a transfer via (e.g., transfer via **508**). In some embodiments, each of the plurality of facets **114** include a seal groove **320** disposed about the chamber openings **304** and configured to house a seal.

(23) The bottom plate **306** and the top plate **308**, and the plurality of facets **114** generally define an interior volume **302** of the link chamber **104**. In some embodiments, the bottom plate **306** includes a central opening **318** for accommodating the transfer robot **106**. In some embodiments, one or more pump ports **108** are disposed through the bottom plate **306**. In some embodiments, the one or more pump ports **108** are disposed radially outward of the central opening **318**.

(24) In some embodiments, the bottom plate **306** includes a plurality of actuator openings **326** proximate corresponding ones of the plurality of chamber openings **304** and configured for passing an actuator of the one or more slit valves **115** therethrough. For slit valves having linear motion actuators, the bottom plate **306** includes actuator openings **326A** that extend substantially parallel with respect to a vertical axis **350** of the link chamber body. For slit valves having “L” motion actuators, the bottom plate **306** includes angled actuator openings **326B** that extend at an angle that is not ninety degrees with respect to the vertical axis **350** of the link chamber body **310**.

(25) In some embodiments, the top plate **308** includes one or more recesses **330** proximate each of the plurality of chamber openings **304**. In some embodiments, the one or more recesses **330** comprise two recesses for each of the plurality of chamber openings **304**. In some embodiments, each of the one or more recesses **330** include fastener openings **332** extending from the one or more recesses **330** to an outer sidewall **338** of the link chamber body **310**. The fastener openings **332** may facilitate coupling the link chamber to chamber components, such as the one or more slit valves **115**, any of the chambers disclosed herein, or a cover plate **410**. In some embodiments, the link chamber body **310** includes lower recesses **354** to facilitate coupling the link chamber **104** chamber components, such as the one or more slit valves **115**, any of the chambers disclosed herein, or a cover plate **410**. In some embodiments, the top plate **308** includes a lid opening **342**. The lid opening **342** may be selectively sealed via a lid (see lid **402**) that is removably coupled to the top plate **308** to facilitate access to the interior volume **302**.

(26) FIG. 4 depicts an isometric view of a link chamber **104** in accordance with at least some embodiments of the present disclosure. In some embodiments, the link chamber **104** includes a lid **402** coupled to the top plate **308** of the link chamber body **310**. In some embodiments, the lid **402** is pivotably coupled to the top plate **308**. In some embodiments, a support apparatus **408** is coupled to the lid **402** and the top plate **308** and facilitates opening and closing of the lid **402**. In some embodiments, the lid **402** includes a handle **428** for manual opening and closing of the lid **402**. In some embodiments, the support apparatus **408** is coupled to the top plate **308** at a first location **432**. In some embodiments, the support apparatus **408** is coupled to the top plate **308** at a second location **434**. In some embodiments, the top plate **308** includes fastener openings **438** at the first location **432** and the second location **434** so that the support apparatus **408** may be coupled to the top plate **308** at various positions as desired.

(27) In some embodiments, the one or more slit valves **115** are coupled to the link chamber body **310** via fasteners **440**, or pins, extending through the fastener openings **332**. In some embodiments, the one or more slit valves **115** includes fastener recesses **470** corresponding with locations of the one or more recesses **330**. In some embodiments, the link chamber **104** includes recess covers **416** configured to cover the one or more recesses **330** and any exposed fasteners or pins. The one or more slit valves **115** may also rest in the lower recesses **354** to further secure the one or more slit valves **115**. Each of the one or more slit valves **115** may include an actuator **422** configured to move a slit valve door **454** to selectively open or close a slit opening **456** for passing a substrate therethrough. In some embodiments, any unused facets of the plurality of facets **114** may be covered with a cover plate **410**.

(28) FIG. 5 depicts a schematic top view of multiple connected multi-chamber processing tools having link chambers in accordance with at least some embodiments of the present disclosure. In some embodiments, the multi-chamber processing tool **100** is coupled to a second multi-chamber processing tool **502** having a second link chamber **504** to facilitate more complex processing. In some embodiments, the second link chamber **504** is similar to or the same as the link chamber **104** discussed above. A plurality of second process chambers **510** may be coupled to the second link chamber **504**. In some embodiments, a second FI **520** is coupled to the second link chamber **504** via one or more second load lock chambers **530**. A second controller **550** may be coupled to the second link chamber **504** and configured to operatively control the second multi-chamber processing tool **502**. In some embodiments, the second controller **550** may be configured to be in communication with the controller **150** of the multi-chamber processing tool **100**. In some embodiments, the controller **150** is configured to operatively control the multi-chamber processing tool **100** and the second multi-chamber processing tool **502**.

(29) In some embodiments, the multi-chamber processing tool **100** is coupled to a second multi-chamber processing tool **502** via a transfer via **508** coupled to one of the chamber openings of the plurality of chamber openings **304** and a corresponding chamber opening of the second link chamber **504**. In some embodiments, more than two link chambers may be similarly coupled together to expand processing capabilities. In some embodiments, the transfer via **508** includes a substrate alignment device **506** that is positioned at one or more of the staging positions. The substrate alignment device **506** is configured to orient a feature on the one or more substrates **112** (i.e., wafer notch) in a correct orientation for processing in the process chambers of the multi-chamber processing tool **100** and the second multi-chamber processing tool **502**. In some embodiments, the transfer via **508** may be angled to advantageously reduce floor space when multiple processing tools are coupled. In some embodiments, the transfer via **508** is coupled to a vacuum pump **560** to pump down the transfer via **508**.

(30) FIG. 6 depicts a schematic top view of a multi-chamber processing tool coupled **100** to a second processing tool **602** in accordance with at least some embodiments of the present disclosure. In some embodiments, the link chamber **104** of the multi-chamber processing tool **100** may be coupled to the second processing tool **602** via the transfer via **508**. In some embodiments, the second processing tool **602** may include the Endura 2 mainframe or an Endura 3 mainframe available from Applied Materials, Inc. of Santa Clara, California. In some embodiments, the second processing tool **602** may comprise any suitable mainframe having a plurality of process chambers. The second processing tool **602** may generally include a FI **620** coupled to one or more transfer chambers **608** via one or more load lock chambers **630**. A plurality of process chambers **610** are coupled to the one or more transfer chambers **608**. Substrates may be transferred between the FI **620**, the transfer chambers **608**, and the plurality of process chambers **610** via transfer robots (not shown) disposed in one or more of the FI **620** and the one or more transfer chambers **608**. The one or more substrates **112** may be transferred from the multi-chamber processing tool **100** to the second processing tool **602** via the transfer via **508** for enhanced processing while advantageously minimizing floorspace. The second processing tool **602** may include a second controller **650**

configured to operatively control the second processing tool **602** and be in communication with the controller **150** of the multi-chamber processing tool **100**. In some embodiments, the controller **150** is configured to operatively control the multi-chamber processing tool **100** and the second processing tool **602**.

(31) While the foregoing is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof.

Claims

1. A link chamber for use in a multi-chamber processing tool, comprising: a link chamber body having a plurality of facets extending between a bottom plate and a top plate, wherein at least seven of the plurality of facets have a chamber opening to form a plurality of chamber openings, wherein the plurality of chamber openings are sized to pass a substrate therethrough, and wherein each of the plurality of chamber openings is configured to be coupled to a slit valve, a load lock chamber, a cover plate, a process chamber, or a second link chamber body, wherein the top plate of the link chamber body includes one or more recesses proximate each of the plurality of chamber openings, wherein each of the one or more recesses includes fastener openings extending from a sidewall of the one or more recesses to a radially outer sidewall of the link chamber body.
2. The link chamber of claim 1, further comprising one or more slit valves coupled to the link chamber and configured to selectively seal one or more of the plurality of facets.
3. The link chamber of claim 1, further comprising a transfer robot disposed in an interior volume of the link chamber body and coupled to the bottom plate.
4. The link chamber of claim 1, further comprising one or more pump ports disposed through the bottom plate.
5. The link chamber of claim 4, further comprising a pump coupled to the one or more pump ports.
6. The link chamber of claim 1, wherein the bottom plate of the link chamber body includes a plurality of actuator openings proximate corresponding ones of the plurality of chamber openings and configured for passing an actuator of a slit valve therethrough.
7. The link chamber of claim 1, wherein the one or more recesses comprise two recesses.
8. The link chamber of claim 1, wherein the top plate includes a lid opening, and further comprising a lid removably coupled to the top plate to close the lid opening, wherein the lid is pivotably coupled to the top plate.
9. A multi-chamber processing tool, comprising: a factory interface (FI) having one or more loadports configured to receive a substrate; the link chamber of claim 1 coupled to the FI; a plurality of process chambers coupled to the link chamber via respective ones of the plurality of chamber openings; one or more load lock chambers coupled to the link chamber via respective one or more of the plurality of chamber openings; and a link robot disposed in the link chamber and configured to facilitate transferring the substrate from the FI to the plurality of process chambers via the one or more load lock chambers.
10. The multi-chamber processing tool of claim 9, wherein the plurality of process chambers include four deposition chambers and two cleaning chambers.
11. The multi-chamber processing tool of claim 9, further comprising a transfer via coupled to one of the plurality of chamber openings, wherein the transfer via is connectable to a transfer chamber of a second processing tool.
12. The multi-chamber processing tool of claim 9, wherein the plurality of process chambers include four process chambers, and the one or more load lock chambers include two load lock chambers.
13. The multi-chamber processing tool of claim 9, wherein the one or more load lock chambers include one load lock chamber, and wherein the one load lock chamber includes a substrate transfer mechanism configured to move the substrate from a location proximate the FI to a location

proximate the link chamber.

14. The multi-chamber processing tool of claim 13, wherein the substrate transfer mechanism is a substrate carousel, a load lock transfer robot, or a servo-controlled shutter.

15. A multi-chamber processing tool, comprising: a factory interface (FI) having one or more loadports configured to receive a substrate; the link chamber of claim 1 coupled to the FI, wherein the bottom plate of the link chamber includes one or more pump ports, and wherein the bottom plate includes a plurality of valve openings proximate corresponding ones of the plurality of chamber openings; a plurality of process chambers coupled to the link chamber via respective ones of the plurality of chamber openings; one or more load lock chambers coupled to the link chamber via respective one or more of the plurality of chamber openings; and a link robot disposed in the link chamber and configured to facilitate transferring the substrate from the FI to the plurality of process chambers via the one or more load lock chambers.

16. The link chamber of claim 6, wherein the plurality of actuator openings are at least partially disposed radially inward of sidewalls defined by the plurality of chamber openings.

17. The link chamber of claim 6, wherein at least one of the plurality of actuator openings extends at an angle that is not ninety degrees with respect to a vertical axis of the link chamber body.

18. The link chamber of claim 1, wherein each of the plurality of facets includes a seal groove disposed about the plurality of chamber openings.

19. The link chamber of claim 1, wherein the link chamber body has an octagonal shape.

20. The link chamber of claim 19, wherein the link chamber body includes seven facets and an eighth side that does not include a chamber opening of the plurality of chamber openings.
